

The Collapse–Churn Model of Emergence

Geometry-First Collapse, Anchoring, and Emergent Physics

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1 Introduction

Standard quantum mechanics treats superposition as ontic and collapse as a probabilistic projection often tied to observation. Relativity treats space and time as geometric backgrounds shaped by energy. These frameworks remain disconnected and struggle with dark matter, dark energy, and the origin of time.

We unify these domains by reinterpreting collapse as the primary physical process. Energy always collapses; persistence emerges only via *anchoring*. Space, time, and matter arise statistically from successful persistence. This *collapse–churn* picture reframes quantum indeterminacy and relativistic geometry as consequences of the same substrate dynamics.

Abstract

We propose a set of postulates unifying quantum measurement, relativity, and cosmology under a single dynamical principle: *continuous collapse and recoherence of energy*. Collapse is not observer-induced but universal and timeless. Persistence arises only when collapses mutually anchor, yielding emergent space, time, and stable matter. Observation is redefined as an emergent form of anchoring, possible only after initial collapses have persisted. Time dilation, gravity, and dark matter follow naturally as statistical consequences of collapse–recoherence churn. The framework is specified via CSL-type dynamics with causal rate modulation by an anchor field, a geometry-first postulate, retarded Green’s functions, and persistence thresholds, leading to falsifiable astrophysical and laboratory predictions.

2 Postulates

Postulate 1: Universal Collapse

All energy collapses continuously and universally into configurations permitted by its probability structure. State evolution follows a CSL-type Itô SDE:

$$d|\psi_t\rangle = \left[-\frac{i}{\hbar} H dt - \frac{1}{2} \int d^3x \Gamma(x) (\hat{A}(x) - \langle \hat{A}(x) \rangle_t)^2 dt + \int d^3x (\hat{A}(x) - \langle \hat{A}(x) \rangle_t) dW_t(x) \right] |\psi_t\rangle, \quad (1)$$

with white-noise correlations

$$\mathbb{E}[dW_t(x)dW_t(y)] = \Gamma(x) \delta^{(3)}(x - y) dt, \quad (2)$$

and *smeared* mass–density collapse operators

$$\hat{A}(x) = \int d^3y g_\sigma(x - y) \hat{M}(y), \quad g_\sigma(r) = \frac{1}{(2\pi\sigma^2)^{3/2}} e^{-|r|^2/(2\sigma^2)}. \quad (3)$$

The local collapse rate is modulated by an anchor field $I(x)$:

$$\Gamma(x) = \gamma_0 [1 + \lambda I(x)], \quad \gamma_0 > 0, \lambda \geq 0. \quad (4)$$

If anchoring fails, collapses immediately recohere. Collapse frequencies follow the Born measure $|\alpha|^2$ (Gleason noncontextuality and quadratic weighting).

3 Collapse dynamics (v1.1)

Averaging (1) yields the Lindblad master equation:

$$\dot{\rho} = -\frac{i}{\hbar} [H, \rho] - \frac{1}{2} \int d^3x \Gamma(x) [\hat{A}(x), [\hat{A}(x), \rho]]. \quad (5)$$

This is completely positive and trace-preserving. The anchor coupling (4) modifies rates only, preserving the quadratic structure and the Born rule.

Off-diagonal suppression. Let $\hat{A}_f = \int f(x) \hat{A}(x) d^3x$ with eigenbasis $\{|a\rangle\}$. Off-diagonals obey

$$\frac{d}{dt} \rho_{ab} \sim -\frac{1}{2} \int d^3x \Gamma(x) (A_a(x) - A_b(x))^2 \rho_{ab} \quad (a \neq b). \quad (6)$$

Pointer selection is toward (smeared) mass density eigenprojectors.

Norm/positivity and no-signaling. With local Hermitian Lindblad densities $L_x = \sqrt{\Gamma(x)} \hat{A}(x)$ and $[L_x, L_y] = 0$ for spacelike separation, microcausality holds. If Γ depends only on past-light-cone data via I propagated by G_{ret} , parameter independence is enforced and no controllable FTL arises.

Postulate 2: Persistence Requires Anchoring

A collapse becomes persistent (*blue*) only when constrained by simultaneous interactions with other collapses or existing persistent structures. Define an *anchor score*:

$$\mathcal{A}(x, t) = \int d^3y G_{\text{ret}}(x - y) \rho_{\text{blue}}(y, t), \quad (7)$$

with causal kernel

$$G_{\text{ret}}(x - y) = \frac{\delta(t_x - t_y - |\mathbf{x} - \mathbf{y}|/c)}{4\pi|\mathbf{x} - \mathbf{y}|}, \quad (8)$$

and persistence condition $\mathcal{A}(x, t) \geq \Theta$ (dimensionless threshold). *Causality*: anchoring influence propagates at $\leq c$ via G_{ret} , enforcing parameter independence (no controllable superluminal signaling).

4 Anchor score formalism (v1.1)

Let $\rho_{\text{blue}}(x, t)$ be the density of persistent collapses; define shadow density $\rho_{\text{shadow}} = \rho_{\text{tot}} - \rho_{\text{blue}}$ (all attempts). Coupled rate equations:

$$\partial_t \rho_{\text{blue}} = \Gamma_{\text{persist}} \rho_{\text{shadow}} - \Gamma_{\text{recohere}} \rho_{\text{blue}}, \quad (9)$$

$$\partial_t \rho_{\text{shadow}} = -\Gamma_{\text{persist}} \rho_{\text{shadow}} + \Gamma_{\text{recohere}} \rho_{\text{blue}}, \quad (10)$$

conserve ρ_{tot} .

Rates from anchoring. Using $\mathcal{A}(x, t)$ and $I(x, t)$:

$$\Gamma_{\text{persist}}(x, t) = \gamma_0 [1 + \lambda I(x, t)] H(\mathcal{A}(x, t) - \Theta), \quad (11)$$

$$\Gamma_{\text{recohere}}(x, t) = \gamma_0 [1 - \lambda I(x, t)] H(\Theta - \mathcal{A}(x, t)), \quad (12)$$

with H the Heaviside step and Θ the persistence threshold (dimensionless). High anchor score $\Rightarrow$ persistence dominates; low score $\Rightarrow$ recoherence dominates.

Shadow fingerprints. Transient failures contribute to effective stress–energy:

$$T_{\text{eff}}^{\mu\nu} = T_{\text{blue}}^{\mu\nu} + \sigma T_{\text{shadow}}^{\mu\nu}, \quad 0 < \sigma \leq 1. \quad (13)$$

Anchor field dynamics. Causal, damped wave equation sourced by blue density:

$$\square I(x, t) + \alpha \partial_t I(x, t) = \kappa \rho_{\text{blue}}(x, t), \quad (14)$$

with damping α and coupling κ .

Postulate 3: Emergence of Space and Time (Geometry-First)

Geometry is the first result of every collapse: each collapse yields a concrete geometric relation (point, distance, or orientation) before any higher-order structure. **Space** is the relational web of these geometric outputs stitched by persistent collapses. **Time** is the ordered record of successive persistent geometric outcomes. The *substrate* is the Hilbert space of possible configurations; “timeless” means ordering arises only from collapse events.

Postulate 4: Recoherence and Stability

Unanchored collapses recohere. Persistent states continually attempt to recohere but are immediately re-collapsed by their anchoring network. Persistence is a dynamic equilibrium.

Postulate 5: Emergent Physics

Our observed world is the persistent residue of churn. Quantum interference is the statistics of persistence vs. recoherence. Time dilation is a bookkeeping load effect. Gravity emerges from anchored networks curving the effective spacetime. Transient collapses act as shadow mass.

Postulate 6: Conservation Through Churn

Define blue/shadow energy densities $\varepsilon_{\text{blue}}, \varepsilon_{\text{shadow}}$ and currents $\mathbf{j}_{\text{blue}}, \mathbf{j}_{\text{shadow}}$ (Sec. 5). Local continuity:

$$\partial_t(\varepsilon_{\text{blue}} + \varepsilon_{\text{shadow}}) + \nabla \cdot (\mathbf{j}_{\text{blue}} + \mathbf{j}_{\text{shadow}}) = 0. \quad (15)$$

5 Conservation law (v1.3)

5.1 Global energy rate from CSL

From (5):

$$\frac{d}{dt}\langle H \rangle = \text{Tr}(H\dot{\rho}) = -\frac{1}{2} \int d^3x \Gamma(x) \text{Tr}(H[\hat{A}, [\hat{A}, \rho]]) = -\frac{1}{2} \int d^3x \Gamma(x) \langle [\hat{A}, [H, \hat{A}]] \rangle. \quad (16)$$

Churn-neutrality: if $[\hat{A}, [H, \hat{A}]] = 0$ (e.g. appropriate smearing and H class), then $d\langle H \rangle/dt = 0$ in isolation. Otherwise the rate is bounded $O(\gamma_0)$ (no GRW-style blow-up) for standard CSL parameters.

5.2 Local continuity

Define

$$\varepsilon_{\text{blue}} = \bar{m}c^2 \rho_{\text{blue}}, \quad \varepsilon_{\text{shadow}} = \sigma \bar{m}c^2 \rho_{\text{shadow}}, \quad \mathbf{j}_{\text{blue}} = \mathbf{v}_{\text{blue}} \rho_{\text{blue}}, \quad \mathbf{j}_{\text{shadow}} = \mathbf{v}_{\text{shadow}} \rho_{\text{shadow}}, \quad (17)$$

where drift fields $\mathbf{v}_{\text{blue,shadow}}$ are induced causally via (14). Multiply (9)–(10) by energy weights and add; choosing $\sigma = 1$ (or enforcing $\Gamma_{\text{recohere}}/\Gamma_{\text{persist}} = \rho_{\text{shadow}}/\rho_{\text{blue}}$) yields the exact local continuity (15):

$$\partial_t \varepsilon_{\text{tot}} + \nabla \cdot \mathbf{j}_{\text{tot}} = 0, \quad \varepsilon_{\text{tot}} = \varepsilon_{\text{blue}} + \varepsilon_{\text{shadow}}, \quad \mathbf{j}_{\text{tot}} = \mathbf{j}_{\text{blue}} + \mathbf{j}_{\text{shadow}}. \quad (18)$$

5.3 Heating bounds

Using (16) for nonrelativistic $H = \sum_i \hat{p}_i^2/2m_i + V(\hat{x})$ and Gaussian smearing,

$$[\hat{A}, [H, \hat{A}]] = \sum_i \frac{\hbar^2}{m_i} |\nabla g_\sigma|^2 * \hat{n}_i(\hat{x}) \Rightarrow \frac{d}{dt} \langle H \rangle \sim \gamma_0 \sum_i \frac{\hbar^2}{m_i \sigma^2} \cdot C(I, \lambda), \quad (19)$$

with $C(I, \lambda) = \int d^3x [1 + \lambda I(x)](\cdots)$ bounded. Standard CSL σ and γ_0 keep heating within experimental limits; anchoring rescales by $(1 + \lambda I) \leq (1 + \lambda)$.

5.4 Diagnostics (for simulation/lab)

Global conservation and causal flux checks:

$$\mathcal{C}_1(t) = \int d^3x \varepsilon_{\text{tot}}(x, t) \text{ (const)}, \quad \mathcal{C}_2(t, \Omega) = \int_\Omega d^3x \varepsilon_{\text{tot}} + \int_{\partial\Omega} \mathbf{j}_{\text{tot}} \cdot d\mathbf{S} = 0. \quad (20)$$

No-signaling test: modulate I in region A; measure $\Delta\rho_{\text{blue}}$ in spacelike-separated B. Expect zero until causal arrival.

6 Relativity mapping (v1.4)

6.1 Weak-field time dilation

Assume local persistence rate scales as $R(x) = R_0 \sqrt{1 - \lambda I(x)}$, giving

$$\frac{d\tau}{dt} = \frac{R}{R_0} = \sqrt{1 - \lambda I(x)} \simeq 1 - \frac{1}{2} \lambda I(x). \quad (21)$$

Match to GR (static weak field $|\Phi|/c^2 \ll 1$):

$$\frac{d\tau}{dt} \simeq \sqrt{1 - \frac{2\Phi}{c^2}} \simeq 1 - \frac{\Phi}{c^2} \Rightarrow \frac{1}{2} \lambda I(x) = \frac{\Phi(x)}{c^2}. \quad (22)$$

Poisson equation relates potential to effective energy density:

$$\nabla^2 \Phi = 4\pi G \frac{\varepsilon_{\text{eff}}}{c^2}, \quad \varepsilon_{\text{eff}} = \varepsilon_{\text{blue}} + \varepsilon_{\text{shadow}}. \quad (23)$$

6.2 Linearized Einstein and PPN

In Lorenz gauge, linearized field equations are

$$\square \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}^{\text{eff}}, \quad g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}. \quad (24)$$

For static isotropic weak fields,

$$ds^2 = -(1 + 2\Phi/c^2) c^2 dt^2 + (1 - 2\gamma_{\text{PPN}} \Phi/c^2) d\mathbf{x}^2. \quad (25)$$

We require $\gamma_{\text{PPN}} = 1 + \delta_\gamma(\lambda, \alpha, \kappa, \sigma)$ with $|\delta_\gamma| \ll 10^{-5}$ (Cassini bound).

6.3 Covariant sketch (Tomonaga–Schwinger)

Define state functional on spacelike hypersurfaces $\Psi[\Sigma]$:

$$\delta\Psi[\Sigma] = \left(-\frac{i}{\hbar} \int_{\Sigma} d\Sigma_{\mu} \hat{T}^{\mu\nu} n_{\nu} - \frac{1}{2} \int_{\Sigma} d\Sigma_{\mu} \Gamma(x) [\hat{A}(x), [\hat{A}(x), \cdot]] \right) \Psi[\Sigma], \quad \Gamma(x) = \gamma_0[1 + \lambda I(x)], \quad (26)$$

with I evolving via a covariant retarded equation $\nabla_{\alpha} \nabla^{\alpha} I = J(\langle \hat{T}^{00} \rangle_{\text{blue}})$. Local densities commute at spacelike separation, preserving microcausality.

6.4 Kinematic+structural dilation

For 3-velocity v ,

$$\frac{d\tau}{dt} = \sqrt{(1 - \lambda I) - \frac{v^2}{c^2}} + O((\Phi/c^2)^2), \quad (27)$$

so kinematic broadening and structural load add at leading order.

7 Relations to existing models

GRW / objective collapse: agrees collapse is objective; extension: continuous and universal, with anchoring determining persistence.

Diósi–Penrose: links gravity and collapse; here collapse is primary, gravity emergent from anchored networks.

Objective Reduction (Orch-OR): collapse intrinsic, not substrate-specific.

Emergent gravity (Jacobson, Verlinde): spacetime emergent; driver is collapse–recoherence rather than entropy.

Page–Wootters: time emergent; arises here from ordered persistent geometry.

Quantum Darwinism: redundancy stabilizes pointer states; *observation* is operationally defined as raising $\mathcal{A}$ above Θ in ≥ 2 subsystems.

Triadic emergence (relational): collapse is binary, but each anchored outcome implies a triad: anchored value, null baseline, inverse reflection. Analogies: Dirac $\pm E$ with vacuum 0, CPT, Pauli triplets, QCD color triads, 3D orientation. Triadic is relational, not three realized outcomes.

8 Methods roadmap

1. **Theoretical development:** Derive decoherence rates including gravitational dilation contribution
2. **Numerical simulation:** Simulate causal automata for I, ρ_{blue} with diagnostics $\{\mathcal{C}_1, \mathcal{C}_2\}$
3. **Observational constraints:** Fit halo profiles and merger lensing to extract $\{\lambda, \Theta, \sigma\}$

4. **Covariant extension:** Develop covariant Tomonaga–Schwinger extension and bound δ_γ

9 Predictions (v1.5)

Key Predictions Summary:

- **Dwarf galaxy suppression:** $M_{\min} \sim 10^{8-9} M_\odot$
- **Small-scale power cutoff:** CMB/Ly- α damping at $k > k_c$
- **Gravitational lensing:** Shadow mass offsets in galaxy mergers
- **Laboratory test:** Decoherence rate shifts $\sim 10^{-15}$ in gravitational fields

9.1 Minimum halo mass $M_{\min}$ (dwarf suppression)

Mean anchor score within virial radius R_{vir} :

$$\overline{\mathcal{A}}(R_{\text{vir}}) \sim \int_{|\mathbf{r}| \leq R_{\text{vir}}} \frac{\rho_{\text{blue}}(\mathbf{r})}{4\pi|\mathbf{r}|} d^3r \gtrsim \Theta. \quad (28)$$

With $\rho_{\text{blue}} \simeq \rho_{\text{vir}} = \Delta_c \rho_{\text{crit}}$ and $M = \frac{4\pi}{3} \rho_{\text{vir}} R_{\text{vir}}^3$:

$$\overline{\mathcal{A}} \sim \rho_{\text{vir}}^{-1/3} M^{2/3} \Rightarrow M_{\min} \sim (\Theta \rho_{\text{vir}}^{1/3})^{3/2}. \quad (29)$$

Tie $\Theta = \chi \rho_{\text{P1}}/\rho_0$ with ρ_0 a fiducial density; expect suppressed halo mass function below $M_{\min} \sim 10^{8-9} M_\odot$ (calibrated by satellites).

9.2 Small-scale power cutoff k_c (CMB/Ly- α)

Persistence acts as a cold component with small-scale suppression when $\mathcal{A} < \Theta$. Use transfer function

$$P_{\text{churn}}(k) = T^2(k) P_{\Lambda\text{CDM}}(k), \quad T(k) = \frac{1}{1 + (k/k_c)^\nu}, \quad \nu \simeq 2-4. \quad (30)$$

Cutoff from anchor reach R_A solving $\mathcal{A}(R_A) = \Theta$: $k_c \sim 1/R_A$. Predict mild damping at $k \gtrsim k_c$ in CMB lensing and Ly- α .

9.3 Lensing and mergers (Bullet-Cluster type)

Shadow is collisionless: lensing peaks (blue+shadow) offset from X-ray gas in mergers. Offsets constrain $\{\sigma, \lambda, \Theta\}$. Forward-model $T_{\text{eff}}^{00} \rightarrow \Phi \rightarrow \kappa$ -maps.

9.4 PPN parameter γ_{PPN}

Solve linearized field with $T_{\text{eff}}^{\mu\nu}$; enforce $|\delta_\gamma| \lesssim 10^{-5}$, constraining $\{\lambda, \alpha, \kappa, \sigma\}$.

9.5 Optomechanics decoherence shift (lab)

Assume rate scales with redshift:

$$\Gamma(h) = \Gamma_0 \sqrt{1 - \frac{2gh}{c^2}} \simeq \Gamma_0 \left(1 - \frac{gh}{c^2}\right), \quad (31)$$

equivalently using (22) with $\frac{1}{2}\lambda I = \Phi/c^2 = gh/c^2$. For $g \simeq 9.8 \text{ m/s}^2$ and $h \simeq 10 \text{ m}$, $\Delta\Gamma/\Gamma_0 \sim 10^{-15}$. Enhance with tall towers, high- Q levitated spheres, or space-based gradients.

9.6 No-signaling test

Prepare spacelike-separated regions A, B . Modulate I in A ; measure ρ_{blue} in B . Expect null response until $t = |\mathbf{x}_B - \mathbf{x}_A|/c$.

10 Conclusion

Core Insight: Collapse–Churn reframes collapse as the universe’s fundamental activity. Geometry is the first output of collapse; persistence arises from anchoring; observation is emergent anchoring. Space, time, relativity, and quantum behavior arise from the same churn.

The model is **causal** (retarded kernels, microcausality), **conservative** (local continuity), and **predictive** (dwarf suppression, small-scale damping, decoherence shifts). It offers a compact program to connect quantum foundations, gravity, and cosmology within a unified framework of emergent spacetime from fundamental collapse dynamics.

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